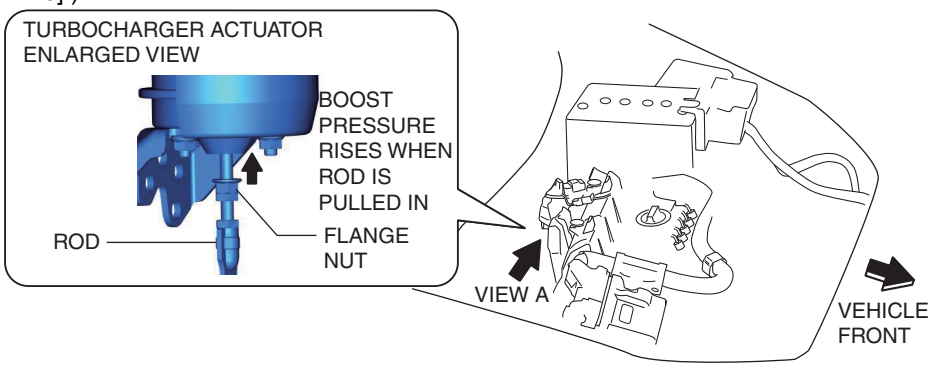
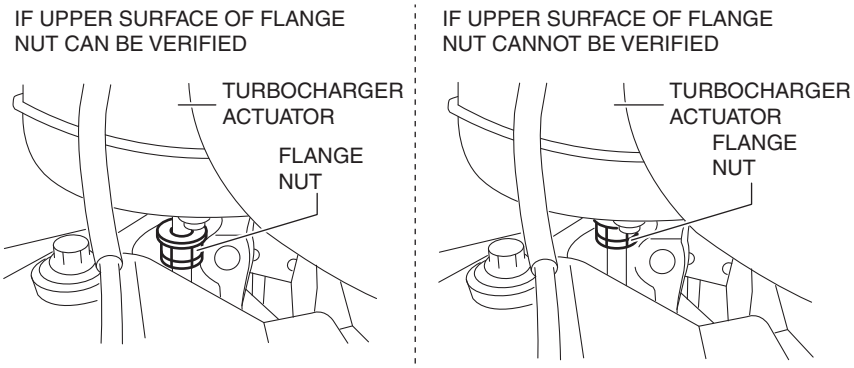
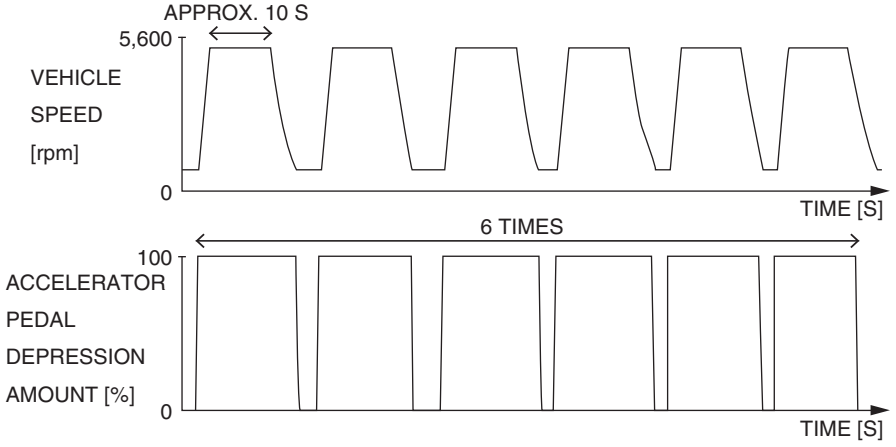
## FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE [SKYACTIV-D 1.5]

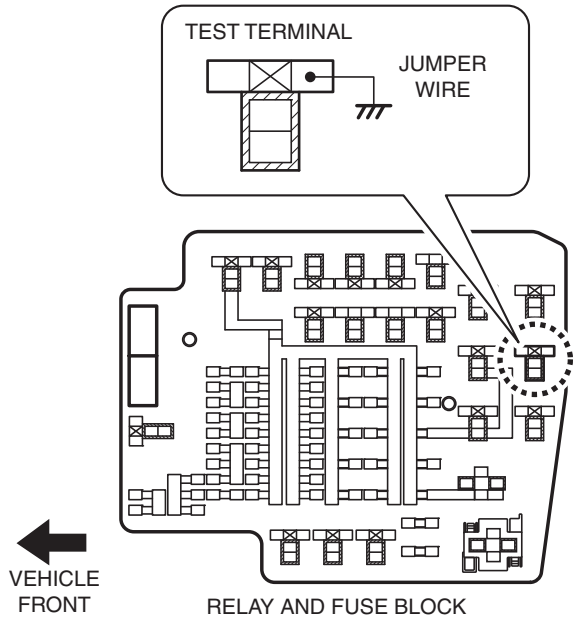
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**Note**
- The carbon deposit on the exhaust valve is removed by performing the procedure for fuel injector carbon deposit burn-off.
- The fuel injector carbon deposit burn-off can be performed when all of the following preconditions are met.
  - — Ambient temperature (servicing area temperature): **0 °C {32 °F} or more** (A/C operation assured)
  - — Intake air temperature: **5 °C {41 °F} or more**
  - — Engine coolant temperature: **75—112 °C {167—233 °F}**
  - — Altitude: **1,500 m or less**
  - — Engine running
  - — The following electrical loads are on:
    - Headlight HI
    - Rear window defogger switch
    - Temperature control dial: MAX HOT
    - Airflow volume control dial: MAX
    - A/C switch
  - — Test terminal is grounded to body

- Perform the following procedures.

| Purpose | Step | Procedure |
|---|---|---|
| Prior preparation | 1 | Verify the engine oil level to prevent overfill and replace the engine oil if it exceeds the X position. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].)<br>• If the engine oil level does not exceed the X position, go to the next step. |
| | 2 | Set the wheel blocks. |
| | 3 | Connect the M-MDS to the DLC-2. |
| | 4 | Switch the ignition ON (engine off). |
| | 5 | Perform the DTC inspection using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• If any DTC other than P1284/P2459 is output, repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].) |
| Engine warming | 6 | Apply the parking brake and verify that the selector lever is in the P position (ATX)/shift lever is in the neutral position (MTX). |
| | 7 | Switch the ignition ON (engine on). |
| | 8 | Turn on the following electrical loads.<br>• Headlight HI<br>• Rear window defogger switch: ON<br>• Temperature control dial: MAX HOT<br>• Airflow volume control dial: 1 step *<br>*: If the ambient temperature is **25 °C {77 °F} or more**, airflow volume control dial can be turned off |
| | 9 | Set the climate control unit to the following.<br>• REC switch (With full-auto air conditioner system): ON<br>• REC/FRESH lever (With manual air conditioner system): ON<br>• MODE dial: HEAT (floor vents) |
| | 10 | • Display the engine coolant temperature (PID: ECT) using the PID/data monitor function. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• While verifying the data monitor item ECT using the M-MDS, run the engine speed at a constant **2,500 to 3,000 rpm** until the engine coolant temperature is **80 °C {176 °F} or more** and warm up the engine. |
| | 11 | Release the accelerator pedal and wait for **10 min or more** while idling. |

| Purpose | Step | Procedure |
|---|---|---|
| Engine warming | 12 | Perform the following procedure to configure the turbocharger with variable turbine geometry to not boost.<br>1. Wait **30 s** after the ignition is switched off. If the cooling fan turned on after switching the ignition OFF, verify that it stopped.<br>2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br>3. Disconnect the turbocharger actuator connector. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br>4. Visually inspect the turbocharger actuator rod from view A shown in the figure and verify that the upper surface of the flange nut secured to the rod is visible.<br>— If the upper surface of the flange nut cannot be visually verified, perform the actuator position sensor (turbocharger with variable turbine geometry) inspection. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY INSPECTION [SKYACTIV-D 1.5].)<br>  <br>  |
| | 13 | Switch the ignition ON (engine on). |
| | 14 | Perform the following procedure six times.<br>• Depress the accelerator pedal while the selector lever is in the P position (ATX)/shift lever is in the neutral position (MTX) and after maintaining the engine speed at **5,600 rpm for 10 s**, release the accelerator pedal.<br>  |

| Purpose | Step | Procedure |
|---|---|---|
| Engine warming | 15 | While verifying the data monitor item ECT using the M-MDS, verify that the engine coolant temperature is **80 °C {176 °F} or more**.<br>• If the engine coolant temperature is **less than 80 °C {176 °F}**, warm up the engine until the engine coolant temperature is **80 °C {176 °F} or more**. |
| | 16 | When the engine is warmed up, turn the following electrical loads off.<br>• Headlights<br>• Rear window defogger switch<br>• Temperature control dial<br>• Airflow volume control dial |
| | 17 | Switch the ignition OFF.<br><br>**Note**<br>• Perform the procedure in Steps 18-21 (engine start) quickly to prevent the engine coolant temperature from decreasing |
| Start fuel injector carbon deposit burn-off | 18 | Open the hood. |
| | 19 | **Note**<br>• DTC P1905:00 may be detected by grounding the test terminal to the body. In this case, perform the memory clearing procedure to clear DTC P1905:00 assuredly even though DTC P1905:00 is automatically cleared by disconnecting the test terminal ground. (See CLEARING DTC [SKYACTIV-D 1.5].)<br><br>Ground the test terminal of the following relay and fuse block using a jumper wire.<br>  <br> |
| | 20 | Close the hood. |
| | 21 | **Note**<br>• Close all doors and windows to maintain cabin temperature and A/C load.<br><br>Close all doors and windows. |
| | 22 | Start the engine. |
| | 23 | Verify that the climate control unit is set to the following.<br>• REC switch (With full-auto air conditioner system): ON<br>• REC/FRESH lever (With manual air conditioner system): ON<br>• MODE dial: HEAT (floor vents) |
| | 24 | **Note**<br>• Perform the procedure quickly to prevent the engine coolant temperature from decreasing.<br><br>Turn on the following electrical loads.<br>• Headlight HI<br>• Rear window defogger switch: ON<br>• A/C switch: ON<br>• Temperature control dial: MAX HOT<br>• Airflow volume control dial: MAX |

| Purpose | Step | Procedure |
|---|---|---|
| Start fuel injector carbon deposit burn-off | 25 | Verify that the i-stop warning light (amber) turns off.<br>• If the i-stop warning light (amber) turns on, press and hold the i-stop OFF switch to verify that the i-stop warning light (amber) turns off. |
| | 26 | Press and hold the i-stop OFF switch.<br>Verify that the i-stop indicator alarm is activated and the i-stop warning light (amber) turns on. |
| | 27 | **Caution**<br>• **If the accelerator pedal is depressed without removing the jumper wire when performing fuel injector carbon deposit burn-off again, be aware that compulsory diesel particulate filter regeneration will start. If compulsory diesel particulate filter regeneration starts, wait until it is completed to prevent damage to the diesel particulate filter.**<br><br>When fuel injector carbon deposit burn-off starts, the wrench indicator light flashes and the engine speed increases to **1,750 rpm**.<br>Verify that fuel injector carbon deposit burn-off starts.<br>• If the start of fuel injector carbon deposit burn-off cannot be verified, remove the jumper wire from the test terminal and perform the procedure from Step 6 (engine warm-up) again. |
| | 28 | Get out of the vehicle and close the door. |
| During fuel injector carbon deposit burn-off | 29 | **Caution**<br>• **If the accelerator pedal is depressed without removing the jumper wire when performing fuel injector carbon deposit burn-off again, be aware that compulsory diesel particulate filter regeneration will start. If compulsory diesel particulate filter regeneration starts, wait until it is completed to prevent damage to the diesel particulate filter.**<br><br>**Note**<br>• The wrench indicator light flashes for approx. **4 min** during fuel injector carbon deposit burn-off.<br>• If fuel injector carbon deposit burn-off was unsuccessful after it started, DTC P1284:00 is detected. Repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].)<br><br>Verify that the wrench indicator light flashes while viewing it from outside of the vehicle.<br>• If the engine warning light and master warning light turn on, remove the jumper wire from the test terminal and perform the procedure from Step 6 (engine warm-up) again. |
| Fuel injector carbon deposit burn-off completed | 30 | Verify that the engine speed reaches the idle speed and the wrench indicator light is turned off while viewing it from outside of the vehicle. (See ENGINE TUNE-UP [SKYACTIV-D 1.5].) |
| | 31 | Switch the ignition OFF. |
| | 32 | Open the hood. |
| | 33 | **Note**<br>• If the jumper wire is disconnected, the engine warning light does not turn on and fuel injector carbon deposit burn-off is interrupted. To verify the procedure was completed correctly, verify that the fuel injector carbon deposit burn-off was completed without the jumper wire coming off.<br><br>Verify that the jumper wire did not come off. |
| | 34 | **Note**<br>• Grounding the test terminal to the body could cause DTC P1905:00 to be detected. In this case, perform the memory clearing procedure to make sure that DTC P1905:00 is cleared assuredly even though it is automatically cleared by disconnecting the test terminal ground. (See CLEARING DTC [SKYACTIV-D 1.5].)<br><br>• Remove the jumper wire installed to the relay and fuse block from the test terminal and ground.<br>• Install the relay and fuse block upper cover. |
| | 35 | Close the hood. |
| | 36 | • Switch the ignition ON (engine off).<br>• Display the DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>— If a DTC is displayed, repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].)<br>• Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) |